

 Did You Know?

A Database Management System (DBMS) helps store, organize and manage data efficiently. Almost every modern application such as banks, hospitals and universities uses a DBMS.

 What is Data?

Data is a collection of raw facts and figures. It can be numbers, text, images or videos that have not yet been processed.

 What is a Database?

A Database is an organized collection of related data that can be stored, accessed and managed efficiently.



What is DBMS?

A Database Management System (DBMS) is software that allows users to create, store, retrieve, update and manage databases easily and securely.



Why do we use DBMS?

- Store large amounts of data
- Retrieve information quickly
- Reduce data duplication
- Improve security
- Maintain data consistency



Components of DBMS

Component	Description
Hardware	Physical devices used to store data.
Software	DBMS application and operating system.
Data	Stored information.
Users	People who access the database.
Procedures	Rules for using the database.



Three-Level Architecture

Level	Purpose
External Level	User View
Conceptual Level	Logical Structure



Simple Example

Student Database



Student ID

Name

Department

CGPA



Applications of DBMS

- Banking Systems
- Hospital Management
- University Systems

- E-commerce Websites
- Airline Reservation
- Library Management

Advantages of DBMS

- Reduces data redundancy
- Improves security
- Data sharing
- Backup and recovery
- Better data consistency

Disadvantages of DBMS

- High cost
- Complex software

- Requires trained users
- Large storage requirement
- Maintenance cost



Interview Questions

Q1: What is DBMS?

DBMS is software used to create and manage databases.

Q2: What is a Database?

A database is an organized collection of related data.

Q3: Name two advantages of DBMS.

Data security and reduced redundancy.

Q4: Name one application of DBMS.

Banking System.

Quick Revision

DBMS Database Management System

Purpose Manage Data

Architecture External, Conceptual, Internal

Main Benefit Organized Data Management